

A Novel Differential Coherent FFH/DS Acquisition Strategy for LEO Satellite-Enabled Internet of Things

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Abstract—With the expanding demands for the space-air-ground integrated Internet of Things (IoT), the low-Earth orbit (LEO) satellite can be regarded as an important complement to IoT networks. Due to the transparency of satellite orbit information and the exposing nature of transmitting links, satellite-ground communication is extremely vulnerable to eavesdropping and jamming attacks. To establish a transmission link, the most crucial procedure was signal acquisition. Existing frequency hopping/direct sequence signal acquisition algorithms were either too time costing for the short visibility window of the LEO satellite or too resource costing for the LEO satellite devices. To alleviate this issue, we propose a novel differential coherent accumulation acquisition strategy for the LEO satellite-enabled IoT network to strike a balance between performance and complexity. It is also demonstrated that the proposed acquisition strategy is capable of achieving high-performance signal acquisition in the low-carrier-to-noise ratio and large dynamic regions. Moreover, we derive and simulate false alarm probability, detection probability, computational complexity, and mean square error of both the delay and Doppler factors in the additive white Gaussian noise channel. Numerical simulation results show that the proposed acquisition strategy improves the performance by 1.6 dB over the noncoherent accumulation strategy but at the expense of 0.43% complexity increase.

Index Terms—Acquisition, differential coherent accumulation, Internet of Things (IoT), low-Earth orbit (LEO) satellite.

I. INTRODUCTION

WITH the rapid development of wireless communication technology, the Internet of Things (IoT) networks have been widely used in various fields of production and life in the modern society [1], [2]. However, it is difficult to achieve the goal of space-air-ground integrated IoT relying on only the existing infrastructure [3]. Satellite network offers service coverage over all-weather and all-terrain, incomparably benefiting the communication requirement of remote

areas [4]. As a key component of the satellite network, the low-Earth orbit (LEO) satellite has the advantages of low-link attenuation, low-transmission delay, and low-launch cost, which is conducive to meeting the requirements of satellite-enabled IoT [5]. Therefore, the LEO satellite can be regarded as a significant complement to the space-air-ground integrated IoT [6]. However, due to the fast growth of node number nodes and openness of wireless networks, LEO satellite-enabled IoT systems suffer severely from both unintentional interference and malicious jamming [7], [8]. The anti-jamming ability of the physical layer is critical.

The commonly used anti-jamming technologies in the physical layer are mainly based on a spread spectrum communication system, including direct sequence spread spectrum (DSSS), frequency hopping spread spectrum (FHSS), and time hopping spread spectrum (THSS) [9]. However, with the development of interference towards high power, large bandwidth, and diversification, the above single anti jamming means are difficult to meet various demands in emerging systems. Hybrid frequency hopping/direct sequence (FH/DS) is a spread spectrum technology composed of DSSS and FHSS. It inherits the merits of both DSSS and FHSS technologies. The FH/DS technology has been widely employed in civilian and military communications with strong anti-jamming ability and low-detection probability. According to the number of symbols transmitted in each frequency hopping slot, FH/DS systems can be classified into two types: 1) slow FH/DS (SFH/DS) and 2) fast FH/DS (FFH/DS). In the SFH/DS system, the FH rate is lower than the symbol rate. On the contrary, the FFH/DS hopping system considers transmitting each symbol via multiple frequency hops, and thus can effectively resist burst errors [10], [11], [12].

The foundation of satellite-ground link establishment is signal acquisition. The acquisition is the procedure in which the existence of the transmitting signal is detected and the channel parameters estimated [13]. For LEO satellite-ground uplinks, due to their inherently short visibility window time [14], the acquisition must be fast enough to guarantee the effectiveness of link establishment and ground user access. On the other hand, since the relative velocity between LEO satellites and ground users varies on a large scale [15], the channel parameters estimation is difficult and resource consuming [16], [17]. However, the LEO satellites are usually small [18], with limited signal processing resources [19].

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So achieving fast FFH/DS signal acquisition based on limited resources was the crucial step for the satellite-ground uplink establishment.

To complete the signal acquisition of LEO satellites, a variety of algorithms have been carried out by researchers. For instance, an acquisition scheme with a recirculation loop has been proposed for the DSSS system in [20]. On this basis, the probability of miss for packet CDMA acquisition has been analyzed in [21]. For higher capacity, a 2-D joint acquisition method for the multicarrier direct sequence code division multiple access systems of the LEO satellite have been described in [13]. In terms of the FH/DS system, a fixed frequency recognition plus DSSS acquisition is used in [22] and [23]. Based on this method, the local carrier frequency scanning methods have been proposed to shorten the acquisition time in [24]. However, the mentioned methods always assume that the number of symbols in each hop are more than one. Therefore, these methods are not suitable for the FFH/DS system.

A handful of acquisition methods for the FFH/DS system have been proposed. Specifically, a parallel down conversion combined with the traditional DSSS acquisition method have been proposed in [25]. But the parallel method greatly increases the receiver complexity. A low-complexity acquisition keystone transform-based method has been described in [26]. Moreover, a rapid serial timing acquisition algorithm for FFH/DS packet radio communication has been presented in [27]. However, none of these methods mentioned above use the energy of multihop, meaning those methods waste most of the energy of symbol and do not fully use the FH diversity gain. A noncoherent accumulation method has been proposed in [28] to gather multihop energy. However, in the case of low-carrier-to-noise ratio (CNR), the loss caused by noncoherent processing is unacceptable. To solve this issue, a coherent-accumulation-based acquisition method has been proposed in [29]. Whereas, coherent accumulation requires a very accurate estimation of the Doppler factor and delay, thereby significantly increasing the computational complexity. In fact, in the DSSS system, in addition to the coherent and noncoherent accumulation, there also exists a differential coherent accumulation technique [30]. Multisymbol energy can be accumulated using differential coherent accumulation in the DSSS system [31]. However, because the energy of a symbol in the FFH/DS system is transmitted at multiple frequency hops, the traditional differential coherence technique of DSSS cannot be used in the FFH/DS system.

This article proposes a novel FFH/DS acquisition strategy based on differential coherent accumulation for LEO satellite-enabled IoT. The inter symbols difference processing has been innovatively introduced to overcome the high-complexity problem of the coherence accumulation acquisition method and the poor sensitivity problem of the noncoherence acquisition method. Such uniqueness is demonstrated in this article by modeling, performance analysis, and simulation comparison. The main contributions of this work are summarized as follows.

- 1) We established the FFH/DS system model considering the impacts of the delay and Doppler effect as well

as the acquisition structure for LEO satellite-enabled IoT. Furthermore, we derived an analytical mathematical expression of the FFH/DS signal before accumulation in detail.

- 2) We developed a novel differential coherent accumulation acquisition strategy for the FFH/DS system. The proposed scheme introduces cross-symbol differential coherent and cross-hop precompensation processing to achieve a high-performance signal acquisition in the low-CNR region.
- 3) Considering the additive white Gaussian noise (AWGN) channels, the statistical characteristics of the proposed differential coherent accumulation results are analyzed. Furthermore, we also derive the explicit mathematical expressions of the false alarm probability, detection probability, and the mean square error (MSE) of the delay and Doppler factor.
- 4) The performance of the proposed algorithm is compared and analyzed via numerical simulations with the existing schemes. The analysis results show that the proposed method improves the performance by 1.6 dB over the noncoherent strategy but at the cost of 0.43% complexity increase.

The remainder of this article is organized as follows. In Section II, the system model and the correlation and Doppler search method before the accumulation of the FFH/DS system are presented. Then, in Section III the proposed acquisition strategy based on differential coherent accumulation is discussed in detail. The theoretical formula of statistical characteristics, the false alarm probability, and the detection probability of the proposed acquisition method are derived in Section IV. And the MSE of the delay and Doppler factor is also analyzed in Section IV. Furthermore, the numerical results are given in Section V followed by the conclusion in Section VI.

Notations: Throughout this article, $\mathbb{R}^{M \times N}$ and $\mathbb{C}^{M \times N}$ stand for the $M \times N$ real- and complex-valued spaces, respectively. Without other specifications, the light-faced, bold-face lower-case, and bold-faced uppercase letters, like c , $\mathbf{c}$, and $\mathbf{C}$, stand for scalars, vectors, and matrices, respectively. For vector $\mathbf{c}$, $\mathbf{c}[l]$ denotes its l th element. $(\cdot)^*$, $(\cdot)^T$, and $\mathcal{F}(\cdot)$ represent the operator for conjugate, transpose, and fast Fourier transform (FFT). Finally, $j = \sqrt{-1}$.

II. SYSTEM MODEL

The architecture of the LEO satellite-enabled integrated IoT is shown in Fig. 1. With the assistance of the LEO satellite, the services in areas that are difficult to cover like the sea, air, and remote mountainous can be implemented more easily. According to Fig. 1, it can be seen that with the increasing densification of various nodes, there are various unintentional interference or malicious jamming from external attackers in the LEO-ground channel. The FFH/DS technology can be used to meet the requirement of anti jamming and anti interception for the LEO satellite. The acquisition of the FFH/DS system is rather critical due to the high-relative movement and the low CNR between LEO satellite and ground devices [13]. The

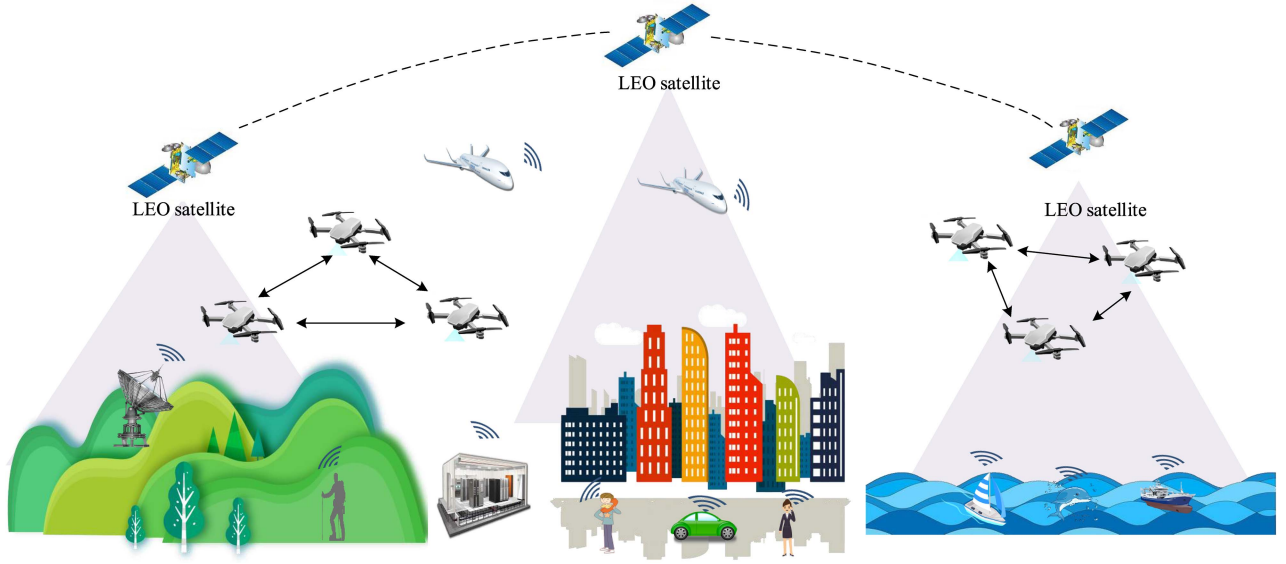


Fig. 1. Architecture of the LEO satellite-enabled IoT.

FFH/DS waveform of the received signal at the LEO satellite is shown in Fig. 2.

The center (radio) frequency of the FFH/DS signal under consideration is assumed to be f_c Hz. For maximum frequency diversity, we supposed in each symbol duration that lasts for T_s seconds, the signal hops onto every one of the K candidate frequencies $\{f_c + f_1, f_c + f_2, \dots, f_c + f_k, \dots, f_c + f_K\}$ for one and only one time. Hence, the dwell on each hop is T_h seconds long where $T_h = T_s/K$. Within any individual hop interval, the signal can be modeled as a fixed-center frequency BPSK DSSS waveform. More particularly, on the k th hop a binary-valued, L -chip pseudo noise (PN) code $c_k \in \{\pm 1\}^{L \times 1}$ is employed to spread the signal's instantaneous bandwidth to $(1 + \gamma)R_c$ Hz, where γ represents the frequency domain roll-off factor of the chip pulse $h(t)$ and R_c denotes the chip rate with its reciprocal $T_c = 1/R_c = T_h/L$ being the chip duration. Generally, L is a power of 2, and this case is only considered below. Without loss of generality, we assumed $\{f_k T_s \in \mathbb{N} | k = 1, 2, \dots, K\}$ for ease of modeling the FFH/DS signal.

To facilitate the initial acquisition process, we assumed the transmission is started with a pilot which is known prior to the receiver. The only distinction between the pilot and the normal FFH/DS signal is the former bears no information, or the symbols that modulate the pilot are constantly $+1$ or -1 for the BPSK constellation. Therefore, the pilot exhibits a period that equals to the symbol duration T_s . The T_s -periodicity of the pilot is the foundation of the proposed accumulation scheme as well as its benchmark counterparts, since they all involve integrating the received signal over the entire pilot, which is assumed to span D consecutive symbols in these schemes all involve.

A. k th hop of the d th Symbol Seen by the Receiver

In this section, the mathematical details of the received pilot signal will be further explored. Omitting the random propagation delay, the Doppler effect, and the thermal noise, the

waveform that pertains to the first hop ($k = 1$) of the first symbol ($d = 1$) in the pilot is given by

$$S_{1,1}(t) = L_p C_1(t) e^{j2\pi(f_c + f_1)t} e^{j\theta} \quad (1)$$

where θ is the random initial phase of the RF carrier in the received signal referenced by the receiver's RF local oscillator (LO) and $C_1(t)$ is the hop-specific direct spreading waveform defined as $C_1(t) = \sum_{l=1}^{L-1} c_k [l + 1] h(t - lT_c)$. L_p is the path loss coming from propagation distance. We note the epoch $t = 0$ in the context of (1) denotes the starting point of the ($d = 1, k = 1$)st correlation window (CW), which is marked by the “★” symbol in Fig. 2.

Now for the more relevant case where both propagation delay and Doppler effect are present. Without loss of generality, we suppose $s_{1,1}(t)$ is now τ seconds delayed with respect to the ($d = 1, k = 1$)th CW, as illustrated in Fig. 1, and the normalized Doppler factor seen by the receiver is α , hence $s_{1,1}(t)$ may be expressed as

$$s_{1,1}(t) = L_p C_1[(1 + \alpha)(t - \tau)] \times e^{j2\pi(1 + \alpha)(f_c + f_1)(t - \tau)} e^{j\theta}. \quad (2)$$

The RF and hopping LOs at the receiver strip off the Doppler-unaffected frequencies in (2) and convert it into a complex-valued baseband equivalent

$$r_{1,1}(t) = L_p C_1(t - \tau) e^{j2\pi\alpha(f_c + f_1)t} \times e^{-j2\pi(1 + \alpha)(f_c + f_1)\tau} e^{j\theta}. \quad (3)$$

For simplicity, throughout this article, we assumed the Doppler effect manifests itself mainly by frequency offset and does not affect the chip rate R_c . This assumption translates into the approximation $C_1[(1 + \alpha)(t - \tau)] \approx C_1(t - \tau)$, which has been invoked in arriving at (3) from (2). Through straightforward derivations, (3) can be extended into the following more general form that accounts for arbitrary d and k :

$$r_{d,k}(t) = L_p C_k(t - \tau_{d,k}) e^{j2\pi\alpha(f_c + f_k)t} e^{j\theta_{d,k}} \quad (4)$$

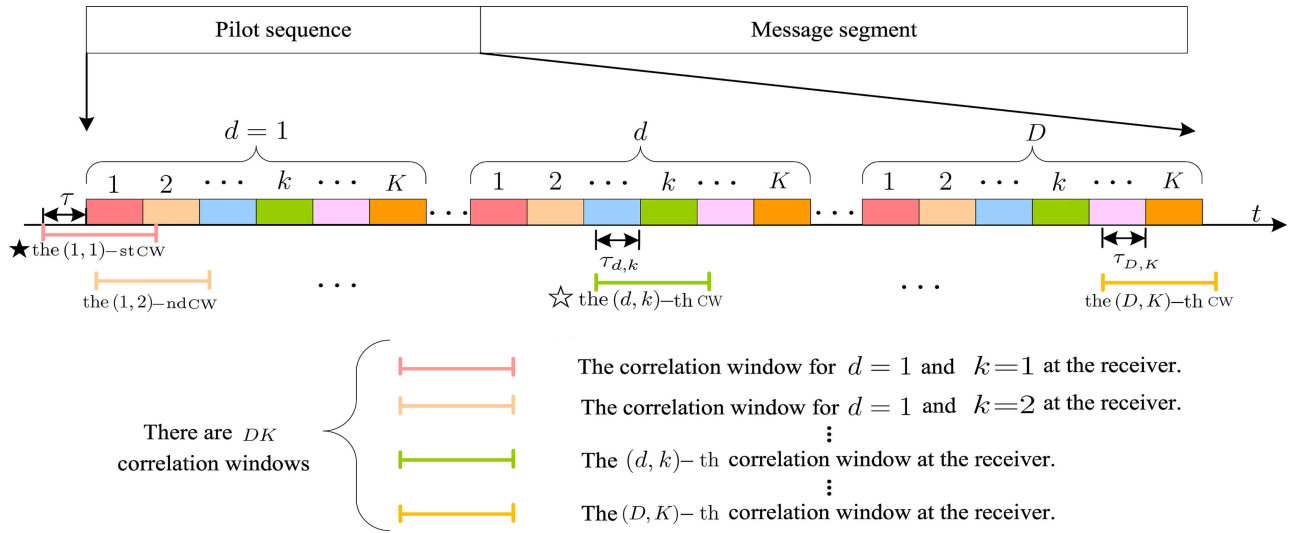


Fig. 2. Waveform of the received signal and the definition of the correlation window.

where

$$\begin{cases} \tau_{d,k} = \tau - \alpha \cdot T_{d,k} \\ T_{d,k} = (d-1)T_s + (k-1)T_h \\ \theta_{d,k} = \theta + 2\pi\alpha f_c T_{d,k} - 2\pi(1+\alpha)f_c \tau \\ -2\pi(1+\alpha)f_k \tau_{d,k}. \end{cases} \quad (5)$$

As in (1) and (3), the epoch $t = 0$ in the context of (4) should reflect the starting point of an associated CW, which is, in this case, the starting point of the (d, k) th CW, as marked by the “ $\star$ ” symbol in Fig. 1. Naturally, (4) will reduce down to (3) if we have $d = k = 1$, as one can easily examine.

B. Correlation and Doppler Search Prior to Cross-Symbol and Cross-Hop Accumulation

The received complex-valued baseband waveform $r_{d,k}(t)$ formulated in (4) shall go through the signal processing procedures illustrated in Fig. 3. At the output of the chip pulse matched filter (MF) therein, we have

$$\tilde{r}_{d,k}(t) = L_p \tilde{C}_k(t - \tau_{d,k}) e^{j2\pi\alpha(f_c + f_k)t} e^{j\theta_{d,k}} \quad (6)$$

where $\tilde{C}_k(t - \tau_{d,k}) = C_k(t - \tau_{d,k}) * h(t)$. The term $e^{j2\pi\alpha(f_c + f_k)t}$ passes the MF almost transparently if the bandwidth of the filter is significantly larger than the residual frequency offset, i.e., if $R_c \gg \alpha(f_c + f_k)$, which is generally the practice case.

Without loss of generality, we assure $\tilde{r}_{d,k}(t)$ is sampled at twice of the chip rate R_c , i.e., $T_{\text{smp}} = (T_c/2)$, which implies the temporal resolution of the propagation delay τ is $(T_c/2)$. As the entire CW has a duration of $2T_h = 2LT_c$ seconds, then the samples within the CW can be found into a $4L \times 1$ complex-valued vector $\mathbf{r}_{d,k}$ as shown in Fig. 3.

Then $\mathbf{r}_{d,k}$ goes through a $4L$ -point FFT and we denote $\mathbf{r}_{d,k} = \mathcal{F}(\mathbf{r}_{d,k}) \in \mathbb{C}^{4L \times 1}$. Before multiplying with the local Fourier transformed PN code template $\mathbf{c}_k \in \mathbb{C}^{4L \times 1}$, $\mathbf{r}_{d,k}$ is directed onto M different cyclic shifting channels in parallel. In the m th channel, $\mathbf{r}_{d,k}$ is cyclically shifted by m bins which amount to compensating the Doppler effect $e^{j2\pi\alpha(f_c + f_k)t}$ in $r_{d,k}(t)$ with a virtual secondary LO $e^{-j(R_h/2)2\pi mt}$, where $R_h = (1/T_h)$.

We referred to $\mathbf{c}_k = [\mathcal{F}(\mathbf{c}_k)]^*$ as local frequency domain PN code template, where $\mathbf{c}_k \in \mathbb{R}^{4L \times 1}$ is a sampled vector of PN code waveforms after padding $2L$ zero samples. More particularly, for $q = 1, 2, \dots, 2L$, We have $\mathbf{c}_k[l] = \mathbf{c}_k(t)|_{t=(l-1)T_{\text{smp}}}$ and $\mathbf{c}_k[l + 2L] = 0$. We note that the length of FFT, IFFT, and $\mathbf{c}_k$ has to be $4L$ to accommodate the CW which has a duration of $2T_h$ seconds. However, as revealed in Fig. 1, the entire uncertainty range of $\tau_{d,k}$ is $[0, T_h)$, only half of the CW's duration. Hence, only 50% of the IFFT outputs are valid correlation data. By discarding the other 50% redundancy, we may obtain the correlation output $\mathbf{a}_{d,k}^m \in \mathbb{C}^{2L \times 1}$ in each of the M channels. Then we form a new matrix $\mathbf{A}_{d,k}$ which encapsulates all the channel specific correlation outputs such that $\mathbf{A}_{d,k} = \{\mathbf{a}_{d,k}^{-(M/2)+1}, \mathbf{a}_{d,k}^{-(M/2)+2}, \dots, \mathbf{a}_{d,k}^{(M/2)}\} \in \mathbb{C}^{2L \times M}$.

Special attention should be paid to $\mathbf{A}_{d,k} = \mathbf{A}_{d,k}[\llbracket n_0, m_0 \rrbracket] = \mathbf{a}_{d,k}^{m_0}[\llbracket n_0 \rrbracket]$, i.e., the n_0 th element of the correlation output in channel m_0 , where

$$\begin{aligned} m_0 &= \underset{m}{\operatorname{argmin}} \left| \frac{R_h}{2} m - \alpha(f_c + f_k) \right| \\ n_0 &= \underset{n}{\operatorname{argmin}} \left| \frac{T_c}{2} n - \tau_{d,k} \right|. \end{aligned} \quad (7)$$

As suggested by the equation above, when $(n = n_0, m = m_0)$ the Doppler effect is (at least coarsely) neutralized, and the PN code in the received pilot gets roughly aligned with the local PN code replica. As a result, $\text{abs}(\mathbf{A}_{d,k})$ shall protrude as a peak out of the background formed by $\{\text{abs}(\mathbf{A}_{d,k}[\llbracket n, m \rrbracket]) | n \neq n_0, m \neq m_0\}$.

This correlation peak plays a vitally important part in all kinds of acquisition schemes. More particularly, through searching for the peak among the background (usually with the assistance of a predefined or adaptive threshold), the receiver decides whether the FFH/DS signal is present or not. And from the 2-D peak position, the receiver extracts initial estimation of the propagation delay and the Doppler factor. In a nutshell, the acquisition performance is directly dictated by the height that the correlation peak could bulge over the background floor. And this, in turn, motivates us

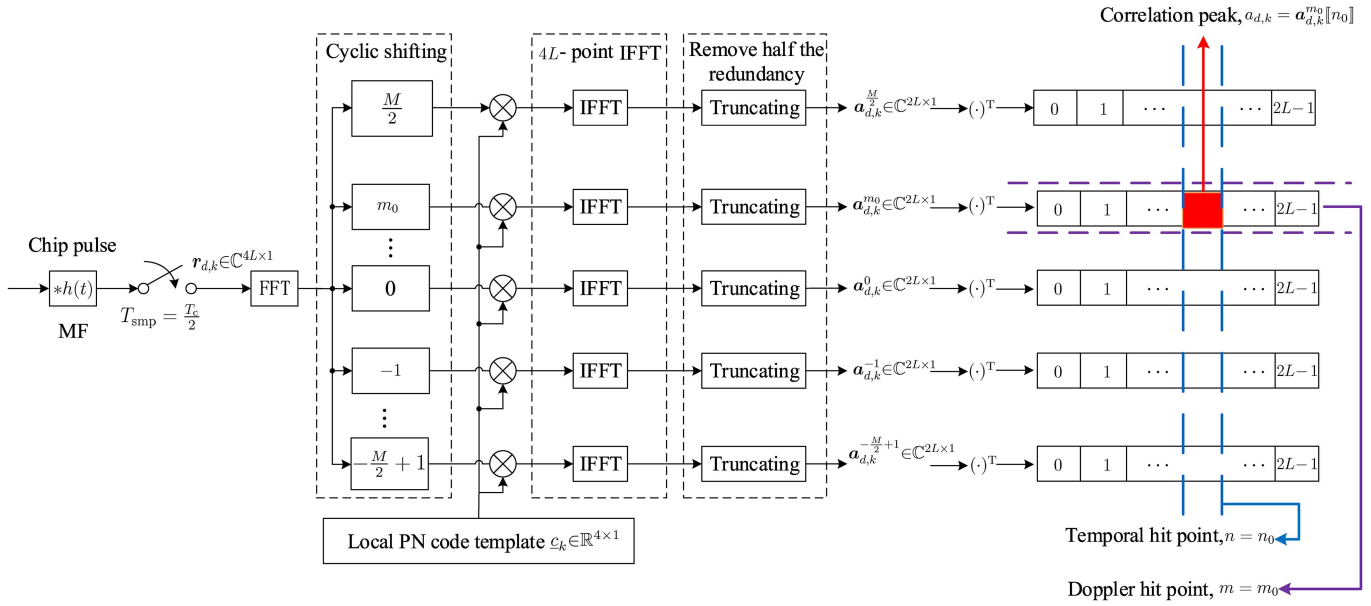


Fig. 3. Block diagram of correlation and Doppler search prior to cross-symbol and cross-hop accumulation.

to use every fraction of the pilot signal. In our context, that amounts to accumulating the one-hop correlation peaks $\{A_{d,k} | 1 \leq d \leq D, 1 \leq k \leq K\}$ both along the index k (cross-hop wise) and along the index d (cross-symbol wise). However, if the phase of $\{A_{d,k} | 1 \leq d \leq D, 1 \leq k \leq K\}$ randomly deviate from one another, then directly combining them together may simply result in a destructive mutual cancellation. To devise an appropriate combining strategy in the following discourse, we will find a concrete expression of $\arg[A_{d,k}]$.

Recalling (7), the Doppler precompensated signal to be fed into the frequency domain cyclic correlator could be formulated as follows (in its continuous form):

$$\tilde{S}(t) = L_p \tilde{C}_k(t - \tau_{d,k}) e^{j2\pi \Delta\alpha(f_c + f_k)t} e^{j\theta_{d,k}} \quad (8)$$

where $\Delta\alpha$ is defined such that $\Delta\alpha(f_c + f_k) = \alpha(f_c + f_k) - (R_h/2)m_0$. Note that the effective correlation interval between $\tilde{S}(t)$ and the local PN code waveform is $[\tau_{d,k}, \tau_{d,k} + T_h]$, therefore the phase of the correlation peak is given by

$$\begin{aligned} \arg[A_{d,k}] &= 2\pi \Delta\alpha(f_c + f_k) \left(\tau_{d,k} + \frac{T_h}{2} \right) \\ &\quad - 2\pi(1 + \alpha)(f_c + f_k)\tau \\ &\quad + 2\pi(1 + \alpha)\alpha f_k T_{d,k} + 2\pi f_c T_{d,k} + \theta \\ &= -2\pi \Delta\alpha(f_c + f_k)\alpha(d - 1)T_s \\ &\quad + 2\pi\alpha\alpha f_k(d - 1)T_s \\ &\quad + 2\pi\alpha f_c(d - 1)T_s \\ &\quad + 2\pi\alpha f_k(d - 1)T_s + \theta(k) \end{aligned} \quad (9)$$

where $\theta(k)$ is the function of the hop index k , and it is independent of the symbol index d . More particularly, we have

$$\begin{aligned} \theta(k) &= \theta + 2\pi \Delta\alpha(f_c + f_k)\tau \\ &\quad - 2\pi \Delta\alpha(f_c + f_k)\alpha(k - 1)T_h \\ &\quad + 2\pi \Delta\alpha(f_c + f_k) \frac{T_h}{2} \\ &\quad - 2\pi(1 + \alpha)(f_c + f_k)\tau \end{aligned}$$

$$\begin{aligned} &+ 2\pi(1 + \alpha)\alpha f_k(k - 1)T_h \\ &+ 2\pi\alpha f_c(k - 1)T_h. \end{aligned} \quad (10)$$

C. On the Uncertainty Range of τ

For the convenience of defining the signal model and formulation of the acquisition problem, in Fig. 1 we have implicitly restricted τ within a relatively narrow uncertainty range of $[0, T_h)$. However, the extension to a more general case of $\tau \in [0, T_s)$ is quite straightforward. In the above two sections, we have explained how to cover the current uncertainty range $[0, T_h)$ with a bank of DK CWs as illustrated in Fig. 1. Since we have $[0, T_s) = \{[0, T_h) \cup [T_h, 2T_h) \cup \dots \cup [(K - 1)T_h, T_s)\}$, the other $(K - 1)$ constituent T_h -long regions may be covered by $(K - 1)$ CW banks, respectively. And it is easy to find out that every two neighbors of these CW banks are temporally offset by T_h s. To summarize, the cost of broadening the searching range of τ by DK times is extra complexity in the same proportion.

III. PROPOSED SCHEME

According to the above analysis in Section II, it is necessary to accumulate the one-hop correlation peaks both along the index k and the index d . Therefore, the accumulation strategy is critical for FFH/DS acquisition. This section will first briefly describe two existing accumulation methods: 1) noncoherent and 2) coherent, for FFH/DS systems. Then, the novel differential coherent accumulation for FFH/DS systems will be described in detail.

A. Noncoherent Accumulation Strategy

According to (9), the phase of correlation peaks $A_{d,k}$ randomly deviate from one another, so the phase must be compensated before accumulation. As shown in Fig. 4, the noncoherent accumulation is to calculate the squared amplitude of the correlation results to remove the influence of

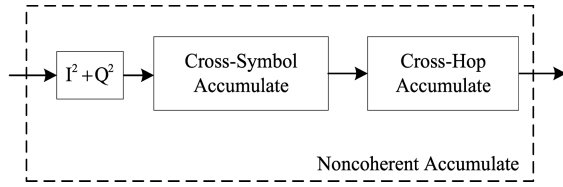


Fig. 4. Block diagram of the noncoherent accumulation strategy.

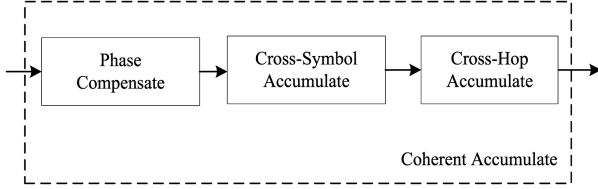


Fig. 5. Block diagram of the coherent accumulation strategy.

phase [38]. Then add the squared amplitude of different hop and symbol correlation results. Thus, the correlation peak accumulation results can be expressed as

$$Y = \sum_{k=1}^K \left[\sum_{d=1}^D \left\{ [\text{Re}(A_{d,k})]^2 + [\text{Im}(A_{d,k})]^2 \right\} \right]. \quad (11)$$

B. Coherent Accumulation Strategy

As shown in Fig. 5, the correlation results of the coherent accumulation acquisition strategy must be accumulated directly without any square and difference operations [39]. Therefore, the Doppler factor and delay must be searched and compensated in a tiny search interval. To redefine the Doppler factor and delay search intervals of coherent accumulation are $R_h/2^N$ and T_c/G . When $R_h/2^N$ is small enough, $\Delta\alpha(f_c + f_k) \approx 0$ and $\alpha \cdot \alpha$ can be ignore. The phase of the correlation peak in (9) can be rewritten as

$$\arg[A_{d,k}] \approx -2\pi(1 + \alpha)(f_c + f_k)\tau + 2\pi\alpha(f_c + f_k)T_{d,k} + \theta. \quad (12)$$

Then the phase compensation value can be expressed as

$$\theta_{\text{comp}} = -2\pi(1 + \hat{\alpha})(f_c + f_k)\hat{\tau} + 2\pi\hat{\alpha}(f_c + f_k)T_{d,k} \quad (13)$$

where $\hat{\alpha}(f_c + f_k) = m_0 R_h/2^N$ and $\hat{\tau} = n_0 T_c/G$. Then, the correlation peak accumulates results based on the coherent strategy can be expressed as

$$Y = \sum_{k=1}^K \left[\sum_{d=1}^D (A_{d,k} \cdot e^{-j\theta_{\text{comp}}}) \right]. \quad (14)$$

C. Proposed Differential Coherent Acquisition Strategy

The noncoherent accumulation strategy utilizes the method of squared correlation peaks to remove the influence of phase. The computational complexity of the noncoherent accumulation strategy is low, but the loss caused by the square is enormous. Coherent accumulation is a maximum likelihood estimation method that can fully use pilot energy but

requires high-computational complexity. Considering the limited resources and the high-performance requirements of LEO satellite, a novel differential coherent accumulation acquisition strategy for coherent FFH/DS hybrid spread spectrum systems have been proposed in this section. Compared with noncoherent accumulation, the proposed differential coherent accumulation improves the acquisition performance with only a little bit of increase in complexity.

The signal model formulated in Section II will be used in the proposed acquisition algorithm. The algorithm contains the following four steps, as illustrated in Fig. 6, *Step 1*: down-conversion; *Step 2*: correlation and Doppler search by FFT; *Step 3*: differential coherent accumulation; *Step 4*: signal detection by threshold test.

1) *Step 1 (Down Conversion)*: According to Section II, before correlation and Doppler search, the RF and hopping LOs must be stripped, i.e., it must be down converted. Equation (4) in Section II have described the down-conversion results of the k th hop of the d th symbol.

2) *Step 2 (Correlation and Doppler Search by FFT)*: As illustrated in Fig. 3 and analyzed in Section II, the second step is correlation and Doppler search by FFT. The specific implementation process of correlation and Doppler factor search has been analyzed explicitly in Section II. The implementation process is shown in Fig. 3, and the correlation results and the correlation peak are defined as $A_{d,k}$ and $A_{d,k}$.

3) *Step 3 (Differential Coherent Accumulation)*: From (9) and (10), the phase of the correlation peak is divided into two parts. The first part only relates to the symbol index d , and the other part only relates to the hop index k .

First, to take the conjugate of the (d, k) th correlation peak and multiply it by the $(d + 1, k)$ th correlation peak, which is called differentially coherent processing across symbols. The phase of the correlation peak after differential coherence can be expressed as

$$\arg[A_{d+1,k} \cdot A_{d,k}^*] = -2\pi\Delta\alpha(f_c + f_k)\alpha T_s + 2\pi\alpha^2 f_k T_s + 2\pi\alpha f_c T_s + 2\pi\alpha f_k T_s. \quad (15)$$

According to the Doppler effect, the Doppler factor α can be expressed as $\alpha = v/c$. v is the relative radial velocity between the receiver and transmitter and c is the speed of light. In the actual situation, the relative radial velocity v is much less than the speed of light c . Therefore, $2\pi\Delta\alpha(f_c + f_k)\alpha T_s$ and $2\pi\alpha^2 f_k T_s$ is so small that it can be ignored. Thus, the phase of the correlation peak after differential coherence processing can be rewritten as

$$\arg[A_{d+1,k} \cdot A_{d,k}^*] = 2\pi\alpha f_c T_s + 2\pi\alpha f_k T_s. \quad (16)$$

The first part of (16) $2\pi\alpha f_c T_s$ is a constant for different symbols and different hops. The second part $2\pi\alpha f_k T_s$ remains the same for different symbols but varies for different hops. Therefore, after differential coherent processing across symbols, the phase of the correlation peak is only related to the hop index k .

To put every hop of the pilot signal into use, the second part $2\pi\alpha f_k T_s$ must be precompensated before cross-hop accumulation. From (7), the estimation of α is $\hat{\alpha} = m_0/2(f_c + f_k)$.

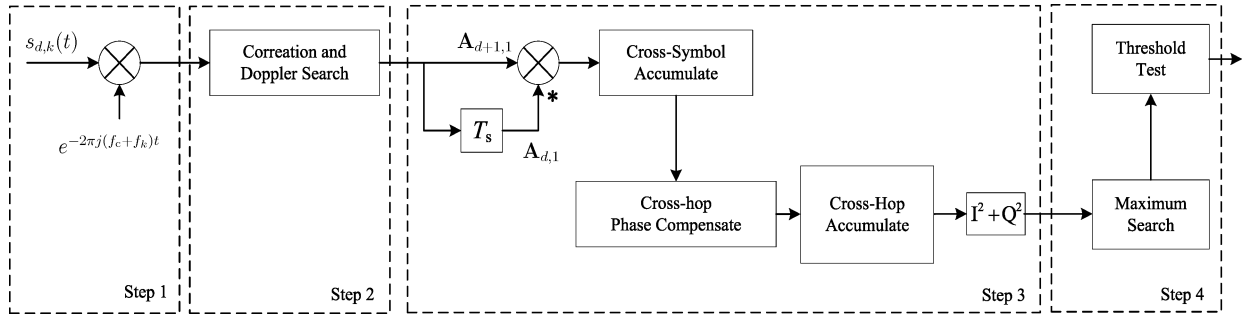


Fig. 6. Block diagram of the proposed differential coherent accumulation acquisition strategy.

Thus, the precompensated phase $\theta_{k,\text{pre}}$ can be expressed as

$$\theta_{k,\text{pre}} = \frac{2\pi m_0}{2(f_c + f_k)} \frac{T_s}{T_h} f_k = \frac{2\pi m_0 K f_k}{2(f_c + f_k)}. \quad (17)$$

According to the above analysis, the correlation peak accumulates results along the index d and index k based on the proposed differential coherent strategy can be expressed as

$$Y = \sum_{k=1}^K \left[\sum_{d=1}^D (A_{d+1,k} \odot A_{d,k}^*) \right] e^{-j\theta_{k,\text{pre}}}. \quad (18)$$

Similarly, all the correlation accumulate results can be expressed as

$$\mathbf{Y} = \sum_{k=1}^K \left[\sum_{d=1}^D (\mathbf{A}_{d+1,k} \odot \mathbf{A}_{d,k}^*) \right] e^{-j\theta_{k,\text{pre}}} \quad (19)$$

where $\odot$ represents the Hadamard product of the matrix and $\mathbf{Y} \in \mathbb{C}^{2L \times M}$.

4) *Step 4 (Signal Detection by Threshold Test):* The square-law detector and single-dwell decision method are used in the proposed differential acquisition strategy. To define $y[n, m]$ to represent the (n, m) th element of the matrix $\mathbf{Y}$, the in-phase and quadrature accumulate results are

$$\begin{cases} y_I[n, m] = \text{Re}[y[n, m]] \\ y_Q[n, m] = \text{Im}[y[n, m]] \end{cases} \quad (20)$$

Calculate the squared amplitude of the element in matrix $\mathbf{Y}$, thus generating a nonnegative matrix $\mathbf{Z} \in \mathbb{N}^{2L \times M}$, where the (n, m) th element is given by

$$z[n, m] = y_I^2[n, m] + y_Q^2[n, m]. \quad (21)$$

Find the maximum value through the matrix $\mathbf{Z}$, i.e., $z[n_0, m_0] = \arg\max_{n,m} z[n, m]$. If the $z[n_0, m_0]$ exceeds the preset threshold, the acquisition is deemed successful. And according to the coordinates (n_0, m_0) of the maximum value, the estimation results of delay $\hat{\tau}$ and Doppler factor $\hat{\alpha}$ are obtained

$$\begin{cases} \hat{\tau} = \frac{T_c}{2} n_0 \\ \hat{\alpha} = \frac{m_0}{2(f_c + f_k) T_h} \end{cases} \quad (22)$$

Otherwise, the acquisition fails and the above steps are repeated to continue the search process.

IV. ACQUISITION PERFORMANCE ANALYSIS

In this section, the statistical characteristics of the differential coherent accumulation results are analyzed first. Then, the theoretical expressions of the probability of detection, P_d , and the probability of false alarm, P_{fa} are developed second. Furthermore, we explored the MSE values of the delay and Doppler factor. Finally, the computational complexity of the proposed method and the existing methods are analyzed.

A. Statistical Characteristics of the Differential Coherent Accumulation Results

The statistical characteristics of the differential coherent accumulation results with noise are analyzed in the section. According to (6), considering the AWGN channel, the complex-valued baseband waveform after sampling with sample rate $T_{\text{samp}} = T_c/2$ can be expressed as

$$\tilde{r}(iT_{\text{smp}}) = L_p \tilde{C}_k(iT_{\text{smp}} - \tau) e^{j2\pi\alpha(f_c + f_k)iT_{\text{smp}}} e^{j\theta_{d,k}} + W(iT_{\text{smp}}). \quad (23)$$

Assuming that the effect of Doppler on spread code can be ignored. This assumption translates into the approximation $\tilde{C}_k(iT_{\text{smp}} - \tau_{d,k}) \approx \tilde{C}_k(iT_{\text{smp}} - \tau)$. And $W(iT_{\text{smp}})$ is zero-mean Gaussian random variables with a variance σ_n^2 .

The process of correlation and Doppler search by FFT can be expressed as

$$a_{d,k}^m(n) = \frac{1}{2L} \text{IFFT} \{ \text{shift}(\text{FFT}[\tilde{r}(iT_{\text{smp}})]) * \underline{\mathbf{c}}_k \} \quad (24)$$

where $m = 0, 1, \dots, M-1$ and $n = 0, 1, \dots, 2L-1$.

By the circumferential correlation theorem, the symbols $R(\cdot)$ is introduced to designate the mean of the correlation results between the input PN code and the receiver local PN code [32], respectively. The $a_{d,k}^m(n)$ can be rewritten as

$$a_{d,k}^m(n) = L_p R(\Delta\tau_n) e^{j\theta_{k,n,m}} \text{sinc}(\Delta f_{m,k}) + W_1(k, n, m) \quad (25)$$

where $\theta_{k,n,m}$ denotes the phase of the complex-valued correlation output.

$W_1(k, n, m)$ is the noise term in the correlation output. $\Delta\tau_n = nT_{\text{smp}} - \tau$ and $\Delta f_m = m/2 - (f_c + f_k)T_h\alpha$.

After accumulating cross symbols, the accumulation results can be expressed as

$$\begin{aligned} & \sum_{d=1}^D \{ a_{d+1,k}^m(n) \} \cdot \{ a_{d,k}^m(n) \}^* \\ &= DL_p^2 R^2(\Delta\tau_n) e^{j\theta_k} \text{sinc}^2(\Delta f_m) + W_2(k, n, m) \end{aligned} \quad (26)$$

where $\theta_k = 2\pi\alpha(f_c + f_k)T_s$ and $W_2(k, n, m)$ is the noise term. After the cross-hop phase compensation and cross-hop accumulation, the accumulation results can be expressed as

$$y[n, m] = DL_p^2 R^2(\Delta\tau_n) \sum_{k=1}^K e^{j\theta_k} e^{-j\theta_{k,\text{pre}}} \text{sinc}^2(\Delta f_m) + W_3(n, m) \quad (27)$$

where $W_3(n, m)$ represents the noise term, according to [33], its variance is given by

$$\begin{aligned} \text{var}[W_3(n, m)] &= KD \left[\frac{\sigma_n^2}{2L} + L_p^2 G(\Delta\tau_n) \text{sinc}^2(\Delta f_m) \right]^2 \\ &+ 2KD L_p^2 R^2(\Delta\tau_n) \left[\frac{\sigma_n^2}{2L} + L_p^2 G(\Delta\tau_n) \text{sinc}^2(\Delta f_m) \right]. \end{aligned} \quad (28)$$

$G(\Delta\tau_n)$ is the variance of the correlation results between the input PN code and the receiver local PN code [32] which can be expressed as

$$G(\Delta\tau_n) = \begin{cases} |\Delta\tau_n|^2/(2L), & |\Delta\tau_n| \leq T_{\text{smp}} \\ (1 - 2|\Delta\tau_n| + 2|\Delta\tau_n|^2)/(2L), & |\Delta\tau_n| > T_{\text{smp}} \end{cases} \quad (29)$$

B. Analysis of False Alarm and Detection Probabilities

Because the input noise is AWGN with zero mean and variance σ_n^2 . The noise term in (27) approximately follows a Gaussian distribution due to the central limit theorem. Thus, assuming H_0 represents the state that the signal is absent. Therefore, the mean and variance of $y_I[n, m]$ and $y_Q[n, m]$ under hypothesis H_0 can be expressed as

$$\begin{cases} \mu_0[y_I] = 0 \\ \mu_0[y_Q] = 0 \\ \sigma_0^2[y_I] = \sigma_0^2[y_Q] = \frac{1}{2}\sigma^2[y] = \frac{KD}{2} \left(\frac{\sigma_n^2}{2L} \right)^2 \end{cases} \quad (30)$$

Similarly, assuming H_1 represents the state that the signal is present. Then, the mean and variance of $y_I[n, m]$ and $y_Q[n, m]$ under hypothesis H_1 can be expressed as

$$\begin{cases} \mu_1[y_I] = DL_p^2 R^2(\Delta\tau_n) \sum_{k=1}^K \text{sinc}^2(\Delta f_m) \sin(\Delta\theta_k) \\ \mu_1[y_Q] = DL_p^2 R^2(\Delta\tau_n) \sum_{k=1}^K \text{sinc}^2(\Delta f_m) \cos(\Delta\theta_k) \\ \sigma_1^2[y_I] = \sigma_1^2[y_Q] = \frac{1}{2}\sigma^2[y] = \frac{1}{2}KDQ_1^2 \\ + \frac{1}{2}KD \left[2L_p^2 R^2(\Delta\tau_n) \text{sinc}^2(\Delta f_m) \right] Q_1 \end{cases} \quad (31)$$

where $Q_1 = [\sigma_n^2/(2L) + L_p^2 G(\Delta\tau_n) \text{sinc}^2(\Delta f_m)]$, $\Delta\theta_k = \theta_k - \theta_{k,\text{pre}}$, and $G(\Delta\tau_n) = [1/(2L)]\varepsilon^2$.

1) *Grid Search for Delay and Doppler Factor:* According to (27), the correlation results are discrete because n and m are integers. As shown in Fig. 7, point O is the actual correlated peak, the position of the highest point of the correlation results on the discrete grid map differs from the ideal highest point due to the quantization error, which can reduce the height of the correlation peak. The actual value of the delay and Doppler factor can be rewritten as

$$\begin{cases} \tau = (n_0 + \varepsilon)T_{\text{smp}} \\ \alpha = (m_0 + \rho) \frac{R_h}{2(f_c + f_k)} \end{cases} \quad (32)$$

where $0 \leq \varepsilon \leq 0.5$ and $0 \leq \rho \leq 0.5$ which can be regarded as the quantization error coefficients of the delay and Doppler factor. In the correlation results, because of the relationship between search resolution and correlation results, there are 16 points

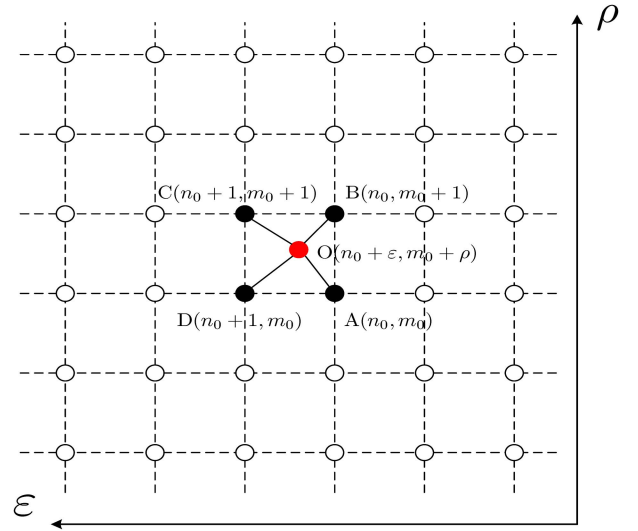


Fig. 7. Diagram of the grid search for delay and Doppler factor.

that can be regarded as candidates for correlation peaks. As shown in Fig. 7, point O is the actual correlated peak. The four discrete points closest to the true value, A(n_0, m_0), B($n_0, m_0 + 1$), C($n_0 + 1, m_0$), and D($n_0 + 1, m_0 + 1$), are considered as the correct estimation position. In the subsequent analysis, we researched the relationship between the detection probability and the Euclidean distance from the correct estimation position A, B, C, and D to the actual position O. Where the Euclidean distance is the Euclidean distance between point O and points A, B, C, and D.

2) *Statistical Characteristics of the Adaptive Threshold:* In this article, the adaptive threshold is usually obtained by multiplying a fixed threshold coefficient γ by the mean of the input noise. The adaptive threshold v [34] can be defined as

$$v = \frac{\gamma}{2LM - 16} \sum_{n=4}^{2L-1M-1} \sum_{m=4} z[n, m] \quad (33)$$

$\{z[n, m] | n = 0, 1, 2, 3; m = 0, 1, 2, 3\}$ are the correlation peak candidates in the correlation results. The adaptive threshold v obeys the central Chi-square distribution with $2(2LM - 16)$ degrees of freedom.

3) *Probability of False Alarm:* Under the hypothesis H_0 , the square-law detector output $z[n, m]$ obeys the central Chi-square distribution (χ^2) with two degrees of freedom. According to [37], its probability density function (PDF) can be expressed as

$$p[z[n, m] | H_0] = \frac{1}{2\sigma_0^2} e^{-\frac{z[n, m]}{2\sigma_0^2}}. \quad (34)$$

Then the false alarm probability can be written

$$\begin{aligned} P_{fa} &= P \left[\max_{n, m} z[n, m] > v; H_0 \right] \\ &= 1 - \{P[z[n, m] < v; H_0]\}^{2LM} \\ &= 1 - \left\{ P \left[\frac{z[n, m]}{v} < \gamma; H_0 \right] \right\}^{2LM} \\ &= 1 - [F(\gamma; H_0)]^{2LM} \end{aligned} \quad (35)$$

where $z[n, m]/v$ obeys the F distribution with degree of freedom of $(2, 2(2LM - 16))$ and $F(\gamma; H_0)$ is the cumulative distribution function of $z[n, m]/v$.

4) *Probability of Detection*: Under the hypothesis H_1 , the correlation peak candidates $\{z[n, m]|n = 0, 1, 2, 3; m = 0, 1, 2, 3\}$ obeys the noncentral Chi-square distribution with two degrees of freedom and noncentral parameter $\lambda_{n,m} = |DB^2 R^2(\Delta\tau_n) \sum_{k=1}^K e^{j\Delta\theta_k} \text{sinc}^2(\Delta f_m)|^2$. According to [35], its PDF can be expressed as

$$p[z[n, m]; H_1] = \frac{1}{2\sigma_1^2} e^{-\frac{z[n, m] + \lambda_{n,m}}{2\sigma_1^2}} I_0 \left[\frac{\sqrt{\lambda_{n,m} z[n, m]}}{\sigma_1^2} \right] \quad (36)$$

where $I_0[\cdot]$ is the zeroth-order-modified Bessel function. And $\{z[n, m]|n = 4, \dots, 2L - 1; m = 4, \dots, M - 1\}$ obeys the central Chi-square distribution with two degrees of freedom. And the PDF is the same as (34).

Without loss of generality, $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$ are assumed the correct estimation positions. The detection probability is the sum of the occurrence probability of the following four events.

- 1) $\max_{n,m} z[n, m] = z[0, 0]$ and $z[0, 0] > v$.
- 2) $\max_{n,m} z[n, m] = z[0, 1]$ and $z[0, 1] > v$.
- 3) $\max_{n,m} z[n, m] = z[1, 0]$ and $z[1, 0] > v$.
- 4) $\max_{n,m} z[n, m] = z[1, 1]$ and $z[1, 1] > v$.

According to [36], [37], the probability of the item 1–4 can be written

$$P_{n,m} = \int_0^{+\infty} p[z[n, m]; H_1] F_v(z[n, m]; H_1) \times \prod_{i=0, i \neq n}^{2L-1} \prod_{j=0, j \neq m}^{M-1} F_{z[i,j]}(z[n, m]; H_1) dz[n, m] \quad (37)$$

where $F_v(x; H_1)$ and $F_{z[i,j]}(x; H_1)$ are the cumulative distribution function of v and $z[i, j]$ under the hypothesis H_1 , respectively.

Therefore, the probability of the detection can be obtained

$$P_d = P_{0,0} + P_{0,1} + P_{1,0} + P_{1,1}. \quad (38)$$

According to (32), the quantization error coefficients of the four correct estimation position are (ε, ρ) , $(1-\varepsilon, \rho)$, $(\varepsilon, 1-\rho)$, and $(1-\varepsilon, 1-\rho)$. Therefore, the probability of the detection can be rewritten as

$$P_d(\varepsilon, \rho) = P_{1,1}(\varepsilon, \rho) + P_{1,2}(\varepsilon, 1-\rho) + P_{2,1}(1-\varepsilon, \rho) + P_{2,2}(1-\varepsilon, 1-\rho). \quad (39)$$

Without loss of generality, assuming ε and ρ are uniformly distributed, i.e., $\varepsilon \sim \mathcal{U}(0, 0.5)$ and $\rho \sim \mathcal{U}(0, 0.5)$. Therefore, the unconditional detection probability can be expressed as

$$\bar{P}_d = \int_0^{0.5} \int_0^{0.5} p_\varepsilon \cdot p_\rho \cdot P_d(\varepsilon, \rho) d\varepsilon d\rho \quad (40)$$

where p_ε is the probability that the quantization error coefficient of the delay is ε . Similarly, p_ρ is the probability that the quantization error coefficient of the Doppler factor is ρ .

TABLE I
ACQUISITION ALGORITHM'S COMPUTATIONAL
COMPLEXITY COMPARISON

Schemes	Computational Cost
Coherent	$(M+2)2^{N+1}DKGL\log_2(2GL) + GLM2^{N+1}(KD+1)$
Proposed	$(M+2)DK2L\log_2(4L) + (D+1)2LKM + 4LKM$
Noncoherent	$(M+2)DK2L\log_2(4L) + D2LKM$

C. Mean Square Error of Delay and Doppler Factor

Based on the above analysis, when the acquisition is successful, the MSE of the delay (normalized by $T_c^2/4$) and Doppler factor (normalized by $1/4$) can be expressed as

$$\begin{cases} \text{MSE}(\hat{\tau}) = \frac{1}{P_d} [\varepsilon^2(P_{0,0} + P_{0,1}) + (1-\varepsilon)^2(P_{1,0} + P_{1,1})] \\ \text{MSE}(\hat{\alpha}) = \frac{1}{P_d} [\rho^2(P_{0,0} + P_{0,1}) + (1-\rho)^2(P_{1,0} + P_{1,1})] \end{cases} \quad (41)$$

Since the probability of detection P_d is the function of (ε, ρ) , to analyze the estimation performance of the proposed acquisition algorithm when ε and ρ varies randomly and simultaneously, the MSE of the Euclidean distance from the correct estimation position $\{(0, 0), (1, 0), (0, 1), (1, 1)\}$ to the actual position (ε, ρ) is developed. For simplicity, we name it the common MSE of delay and Doppler factor which can be expressed as

$$\begin{aligned} \text{MSE}(\hat{\tau}, \hat{\alpha}) &= (\varepsilon^2 + \rho^2) \frac{P_{1,1}}{P_d} + [\varepsilon^2 + (1-\rho)^2] \frac{P_{1,2}}{P_d} \\ &+ [(1-\varepsilon)^2 + \rho^2] \frac{P_{2,1}}{P_d} + [(1-\varepsilon)^2 + (1-\rho)^2] \frac{P_{2,2}}{P_d}. \end{aligned} \quad (42)$$

Then, the common unconditional MSE of delay and Doppler factor can be expressed as

$$\overline{\text{MSE}}(\hat{\tau}, \hat{\alpha}) = \frac{1}{P_d} \int_0^{0.5} \int_0^{0.5} p_\varepsilon \cdot p_\rho \cdot M d\varepsilon d\rho \quad (43)$$

where $M = (\varepsilon^2 + \rho^2)P_{1,1} + [\varepsilon^2 + (1-\rho)^2]P_{1,2} + [(1-\varepsilon)^2 + \rho^2]P_{2,1} + [(1-\varepsilon)^2 + (1-\rho)^2]P_{2,2}$.

D. Complexity Analysis

The complexity of the proposed differential coherent, the existing noncoherent, and the coherent acquisition methods are analyzed in the section. Since the three acquisition strategies all need down conversion and filtering in the early stage, the corresponding calculation complexity is not considered. The complex multiplier consumes much more resources than the complex adder in the actual calculation, so only the calculation times of the complex multiplier are considered here.

Table I shows the computational complexity of the coherent accumulation strategy is the largest, while the computational complexity of noncoherent accumulation is the lowest. The computational complexity of the proposed method is less than that of coherent accumulation and a little bit more than that of noncoherent accumulation.

The computational complexity of three kinds of acquisition algorithms has been shown in Fig. 8. with different L, K, G ,

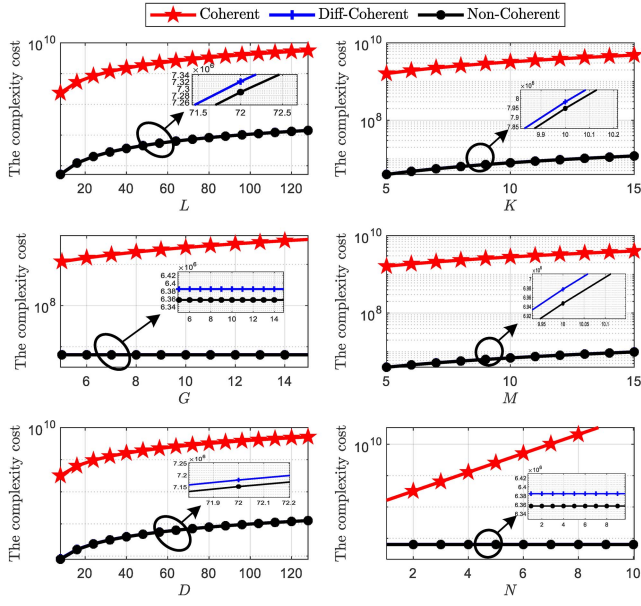


Fig. 8. Computational complexity of three different acquisition methods versus L , K , G , M , D , and N .

M , D , and N . According to Fig. 8, as the increase of L , K , G , M , D , and N , the computational complexity of coherent accumulation strategy increases. And as the increase of L , K , M , and D the computational complexity of differential coherent and noncoherent accumulation strategy increases. The computational complexity of differential coherent and noncoherent accumulation algorithms is independent of N and G . As Fig. 8, the computational complexity of the coherent accumulation strategy is the largest, while the computational complexity of noncoherent accumulation is the lowest. The computational complexity of the proposed method is less than that of coherent accumulation and a little bit more than that of noncoherent accumulation.

V. NUMERICAL ANALYSIS

A. Simulation Settings

In this section, we investigate the detection probability, false alarm probability, and the common MSE of the delay and Doppler factor by theoretical calculation and computer simulations. We also depict the detection probability of the existing coherent and noncoherent strategies, for comparison. Moreover, we also apply the traditional differential acquisition method for the DSSS system [30] to the FFH/DS system and compare its acquisition performance with the counterpart of the proposed acquisition method in this article. Constant false alarm detection is adopted with $P_{fa} = 0.01$ for the simulations of detection probability and MSE. Table II lists the simulation conditions.

B. False Alarm Probability

The false alarm probability P_{fa} of the proposed method against the threshold coefficient γ , is shown in Fig. 9. The threshold coefficient γ varies from 8 to 15. Furthermore, the false alarm probability with the different number of

TABLE II
SIMULATION PARAMETERS

Parameter	Value
DS Processing Gain L	64
Frequency bins number K	8
Doppler coefficient search range M	9
FFH hopping duration T_h	10^{-5} (s)
Number of coherent acquisition delay search cell G	10
Number of coherent acquisition frequency offset search cell N	5

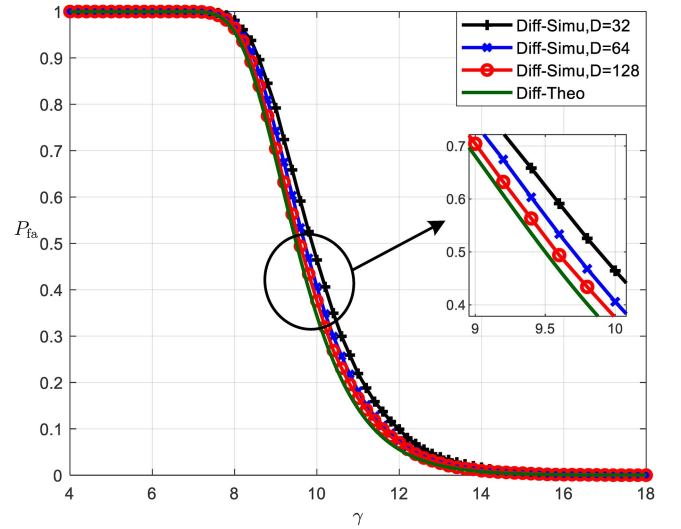


Fig. 9. False alarm probability of the proposed method versus threshold coefficient γ , with the various number of symbols $D = \{32, 64, 128\}$.

accumulated symbols, i.e., $D = \{32, 64, 128\}$, are also shown in the figures for comparison.

From Fig. 9 we can see the false alarm probability decreases as threshold coefficient γ increases. And with the increase of D , the simulation results gradually reach the theoretical result because of the central limit theorem.

C. Detection Probability

The detection probability of the proposed differential coherent accumulation acquisition method in terms of the CNR and the number of symbols D are plotted in Fig. 10. From Fig. 10, the detection probability of the proposed method increases with the increase of the number of symbols D and CNR. The simulation results are in good agreement with the theoretical results.

The detection probabilities of the proposed method with various quantization error coefficients (ε, ρ) , setting as $(0, 0)$, $(0.5, 0)$, $(0, 5)$, $(0.5, 0.5)$ and a random value which obeys uniformly distributed, i.e., $\varepsilon \sim \mathcal{U}(0, 0.5)$ and $\rho \sim \mathcal{U}(0, 0.5)$, are shown in Fig. 11. From Fig. 11, we can see the detection probabilities of the proposed method decrease as the quantization error coefficients increase. When the quantization error coefficients (ε, ρ) are zero, the performance is the best. Conversely, the performance is the worst when the quantization error is the largest. Moreover, when the quantization error

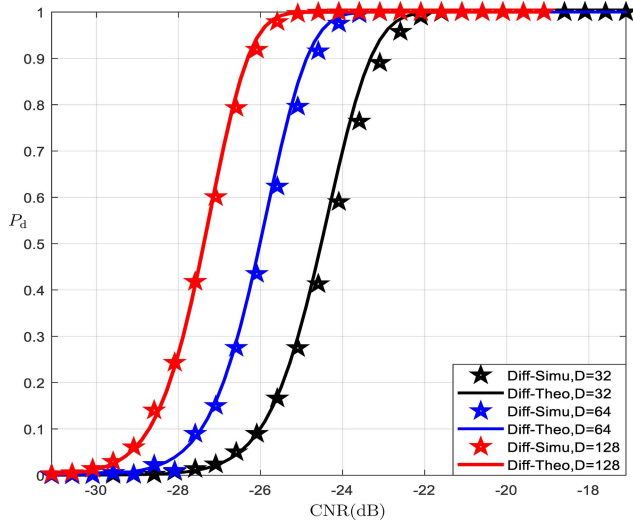


Fig. 10. Detection probability of differential coherent accumulation versus carrier to noise ratio CNR, with the various number of symbols $D = \{32, 64, 128\}$ and fixed $(\varepsilon, \rho) = (0, 0)$.

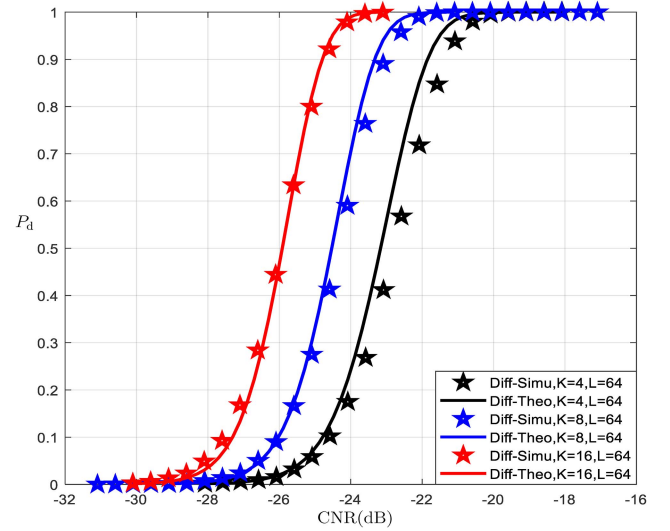


Fig. 12. Detection probability of differential coherent versus CNR, with various number of hop $K = \{4, 8, 16\}$ and the various length of PN code $L = \{128, 64, 32\}$ and fixed $(\varepsilon, \rho) = (0, 0)$.

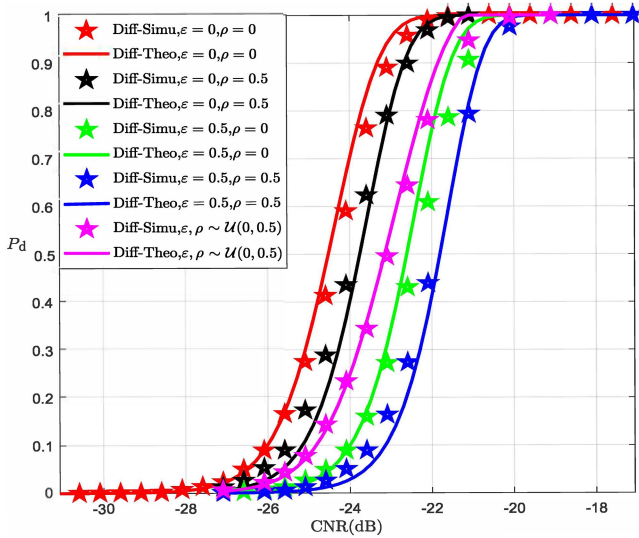


Fig. 11. Detection probability of differential coherent versus CNR, with the various delay and Doppler factor quantization error coefficients, where the $D = 32$ is assumed.

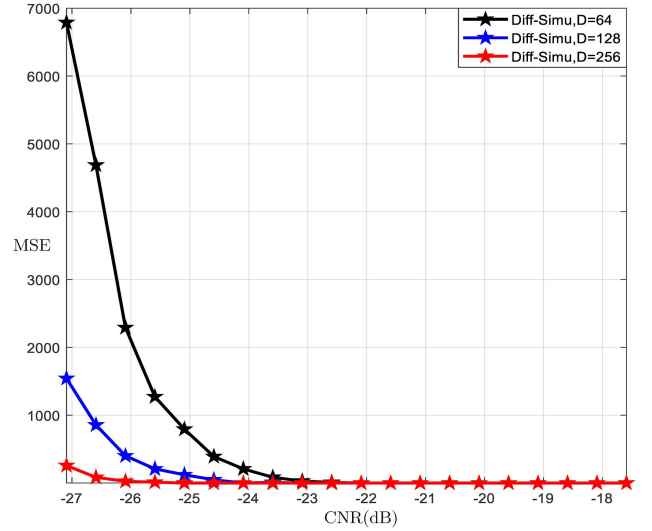


Fig. 13. Common MSE of the delay and Doppler factor when acquisition successfully versus CNR when ε and ρ are all drawn from the uniform $\varepsilon \sim \mathcal{U}(0, 0.5)$ and $\rho \sim \mathcal{U}(0, 0.5)$, with various number of symbols $D = \{64, 128, 256\}$.

is randomly distributed between the maximum and minimum, the performance is in the middle.

Fig. 12 shows the detection probabilities of the proposed method in terms of the CNR and the K , where a fixed $D = 32$ and a fixed false alarm probability $P_{fa} = 0.01$ are assumed. The hop number is set as $K = \{4, 8, 16\}$. Fig. 12 shows that the acquisition performance improved with the increase of K .

D. Common MSE of the Delay and Doppler Factor

The common MSE of the delay and Doppler factor in terms of CNR and accumulation number of symbols are plotted in Figs. 13 and 14. The delay and Doppler factor quantization error coefficient (ε, ρ) are drawn from the uniform $\varepsilon \sim \mathcal{U}(0, 0.5)$ and $\rho \sim \mathcal{U}(0, 0.5)$ with D varying among $\{64, 128, 256\}$.

In engineering applications, the receiver can only judge whether the correlation peak is through the threshold, and cannot know whether the current result is correct, so there are cases of wrong acquisition. Fig. 13 shows the common MSE of the delay and Doppler factor in engineering applications. In the case of low SNR, the probability of wrong acquisition is high, the MSE is large. As CNR increases, the common MSE decreases with decreasing probability of wrong acquisition. When the probability of wrong acquisition gradually reaches zero, the MSE reaches a fixed value. On the premise of a successful acquisition, the common MSE curves of delay and Doppler factor in terms of CNR and the number of symbols are described in Fig. 14. From Fig. 14, the MSE is a convex function of CNR. With the increase of CNR, the MSE of the proposed method increases gradually at first, but after

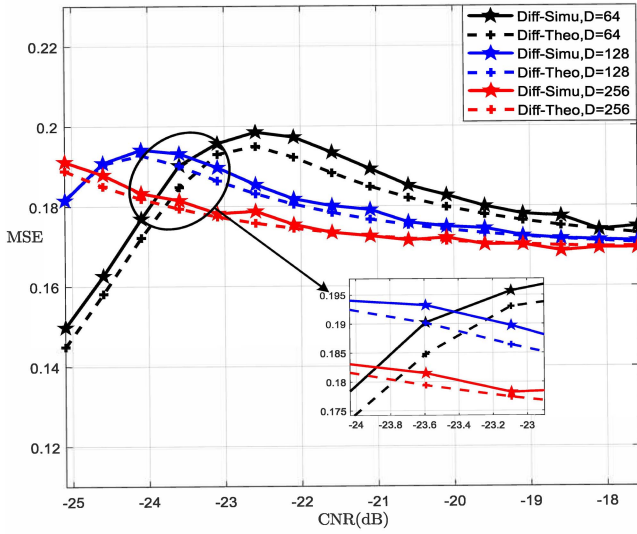


Fig. 14. Common MSE of the delay and Doppler factor when the correlation peak exceeds the threshold versus CNR when ε and ρ are all drawn from the uniform $\varepsilon \sim \mathcal{U}(0, 0.5)$ and $\rho \sim \mathcal{U}(0, 0.5)$, with various number of symbols $D = \{64, 128, 256\}$.

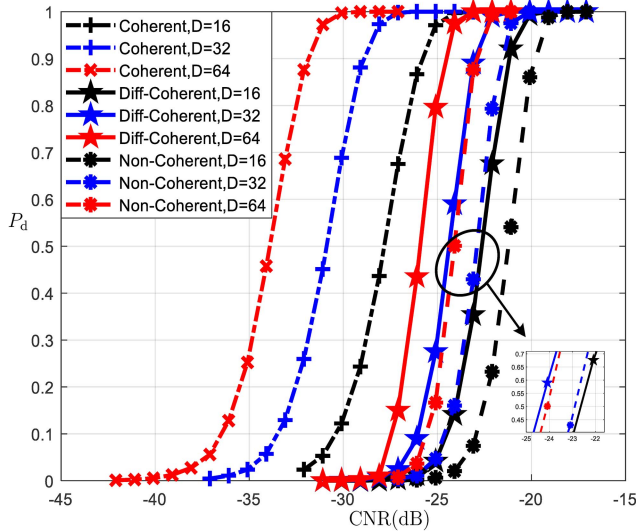


Fig. 15. Detection probability of three different acquisition methods versus CNR, with the various number of symbols $D = \{32, 64, 128\}$ and fixed $(\varepsilon, \rho) = (0, 0)$.

the MSE rises to the maximum value, will decrease gradually, and finally stabilizes to a fixed value. The simulation results are in good agreement with the theoretical results.

E. Comparison of Differential Coherent, Noncoherent, and Coherent Acquisition Strategy

The detection probabilities of three different acquisition methods, differential coherent, coherent, and noncoherent accumulation acquisition methods, with the various numbers of symbols $D = \{32, 64, 128\}$ have shown in Fig. 15. And the computational complexity of three acquisition methods with the various numbers of symbols is shown in Table III. It can be seen that since the coherent accumulation acquisition method is the maximum likelihood estimator, the detection

TABLE III
COMPUTATIONAL COST

Schemes	D = 32	D = 64	D = 128
Coherent	1285306348	2570244056	5140119472
Proposed	3206144	6384640	12741632
Noncoherent	3178496	6356992	12713984

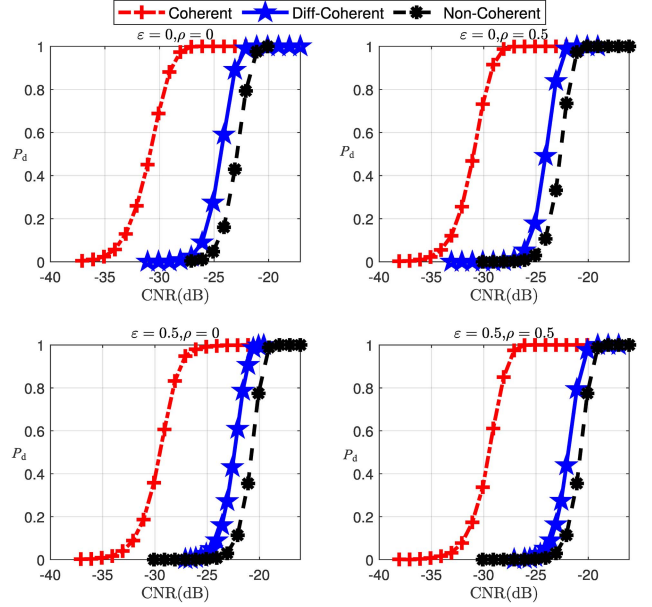


Fig. 16. Detection probability of three different acquisition methods versus CNR, with the various delay and Doppler factor quantization error coefficients and fixed number of symbols $D = 32$.

probability of the coherent accumulation acquisition method is better than the other two methods, while the noncoherent accumulation method performance is the worst. The performance of the proposed differential coherent accumulation method falls between the other two methods. When the number of symbols is 64, the CNR required for the proposed method to achieve 99% detection probability is 1.6 dB lower than that required for the noncoherent accumulation method. In this case, the computational complexity of the proposed method merely increases by 0.43%.

Fig. 16 shows the detection probabilities of coherent, differential coherent, and noncoherent accumulation acquisition methods with various delay and Doppler factor quantization error coefficient (ε, ρ) , setting as $(0, 0)$, $(0, 0.5)$, $(0.5, 0)$, $(0.5, 0.5)$. From Fig. 16, it can be seen that the performance difference between the three acquisition methods is the same with different delays and Doppler factors.

F. Comparison Between the Proposed Differential Method and the Traditional Differential Method

The detection probabilities of the proposed method for the FFH/DS system and the traditional method, with various numbers of symbols $D = \{32, 64, 128\}$ have shown in Fig. 17. The detection probabilities of the proposed method and the traditional differential method both increase as the number of

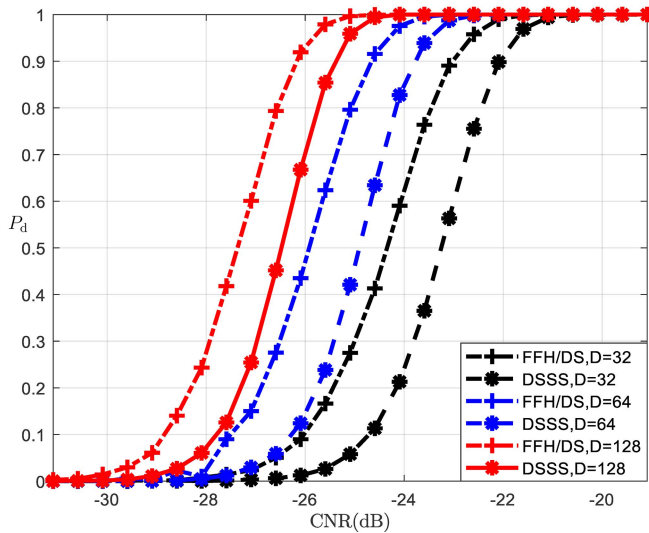


Fig. 17. Detection probability of the proposed method for FFH/DS and the traditional method versus CNR, with the various number of symbols $D = \{32, 64, 128\}$.

symbols D and CNR. It can be seen that the proposed differential method for the FFH/DS system is superior to the traditional differential method for the DSSS system.

VI. CONCLUSION

A realistic FFH/DS system model with delay and Doppler and acquisition structure has been established. Based on the model, a low-complexity differential coherent FFH/DS acquisition strategy for LEO satellite-enabled IoT is proposed in this article. The proposed method achieves high-performance signal acquisition utilizing differential coherent accumulation and precompensation, which can strike tradeoff between calibration complexity and performance. To evaluate the performance of the proposed algorithm, the false alarm probability, detection probability, and the MSE of the delay and Doppler factor are derived and simulated in the AWGN channel. Furthermore, the computational complexity of the proposed method is also analyzed. Compared to the existing acquisition strategies, the proposed scheme improves the performance by 1.6 dB over the noncoherent strategy at the cost of 0.43% complexity increase.

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